

Lab: Branching and Merging in Git

Objective

- Understand branching and merging.
- Create a new branch and merge it with the master branch.

Commands - Branching

1. Create a new branch GitNewBranch

```
$ git branch GitNewBranch
```

- 2) List all local and remote branches

```
$ git branch -a
```

- 3) Switch to the new branch

```
$ git checkout GitNewBranch
```

- 4) Verifying the Status of the Repository

```
$ git status
```

- 5) Commit the changes

```
$ git add.  
$ git commit -m "Added file1.txt in GitNewBranch"
```

Commands – Merging

1. Switch to the master branch

```
$ git checkout master
```

2. View differences between master and branch

```
$ git diff master GitNewBranch
```

3. View visual differences using P4Merge

```
$ git difftool master GitNewBranch
```

4. Merge the branch into master

```
$ git merge GitNewBranch
```

5. View commit history after merging

```
$ git log --oneline --graph --decorate
```

6. Delete the branch and check status

```
$ git branch -d GitNewBranch  
$ git status
```